



Improving Generalization of End-to-End ASR through Diversity and Independence Regularization

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Abstract

Automatic speech recognition (ASR) has been driven by representative end-to-end model architectures, including connectionist temporal classification (CTC), attention-based encoder-decoder (AED), and recurrent neural network transducer (RNN-T). However, these models are prone to overfitting during training, which degrades their generalization performance. In this paper, we propose a novel regularization technique applicable to various ASR models: diversity loss and independence loss. Diversity loss reduces the similarity between feature representations, encouraging the model to learn diverse patterns. Independence loss minimizes the covariance between feature vectors, ensuring that they contain independent information and reducing redundancy. We apply these techniques to CTC, AED, and RNN-T models and demonstrate that the proposed regularization method effectively improves the model generalization performance and robustness through extensive experiments.

Index Terms: speech recognition, diversity loss, independence loss, regularization

1. Introduction

Speech recognition technology has transitioned from traditional GMM-HMM models [1] to modern end-to-end (E2E) neural networks, including connectionist temporal classification (CTC) [2], attention-based encoder-decoder (AED) [3], and recurrent neural network transducer (RNN-T) [4] models. These approaches achieve outstanding performance without intricate preprocessing, yet larger models are susceptible to overfitting [5]. CTC assumes conditional independence and uses a blank token for length mismatches [2], but its lack of explicit alignment restricts complex pattern modeling and context understanding [5, 6]. AED focuses attention effectively but requires vast training data due to its large parameter space [3]. RNN-T supports flexible alignment for online use but can over-optimize noisy or long utterances [4]. Overfitting worsens when internal feature representations fixate too narrowly on specific speech patterns or acoustic conditions. Although speech signals naturally contain repeated phonemes and similar acoustic events, excessive similarity among feature vectors biases the model and undermines generalization. Controlling this redundancy while promoting diverse and independent features remains essential.

To address the overfitting challenge, this paper proposes a diversity and independence regularization strategy for CTC, AED, and RNN-T models. Specifically, two loss terms—diversity loss and independence loss—moderate the cosine similarity and covariance among encoder feature vectors. The diversity loss reduces similarity between feature vectors

within an utterance to broaden the feature representation space, whereas the independence loss reduces covariance between feature vectors to prevent overlap across dimensions. Experiments using our approach on multiple datasets with various noise levels, including conversational speech, demonstrate consistent reductions in word error rate (WER). On KsponSpeech and CHiME-3, our method achieves a notable relative error reduction rate (RERR), underscoring its robustness against challenging acoustic conditions. Moreover, our method introduces minimal computational overhead as it simply introduces two additional loss terms without altering the original network structure. The main contributions of this paper are as follows:

- We propose an architecture agnostic regularization technique based on diversity and independence loss, which integrates seamlessly into various E2E ASR models to mitigate overfitting.
- Our regularization method incurs negligible computational overhead, as it solely introduces additional loss terms during training without altering the model architecture, ensuring efficient and scalable integration.
- We validate the effectiveness of our approach on datasets with high background noise and complex conversational speech, demonstrating improved model robustness and generalization.

2. Related works

Various methods have been proposed to prevent overfitting and stabilize training. Representative techniques include dropout [7] and batch normalization [8], which mitigate excessive reliance on specific paths or channels by regulating parameter variance at the neuron or channel level. Recently, approaches that reduce redundancy among feature vectors to learn richer representations have drawn attention [9, 10], alongside work on regularization strategies aimed at increasing model diversity [11, 12]. For example, VICReg [13] integrates control of feature variance, invariance, and covariance in self-supervised learning to guide models toward diverse representations. Studies have also examined feature diversity and decorrelation in audio representation learning to prevent model collapse [14]. Input-level augmentations such as SpecAugment [15] and Frame-Level SpecAugment [16] similarly target feature redundancy.

In CTC-based ASR architectures, several methods target overfitting and improved training regularization [17–20]. For instance, Mask CTC [21] mitigates overfitting by masking parts of the input sequence, encouraging the model to learn diverse patterns while partially addressing the conditional independence assumption in CTC. Improved Mask-CTC [22] enhances this approach with a conformer-based encoder and dynamic length

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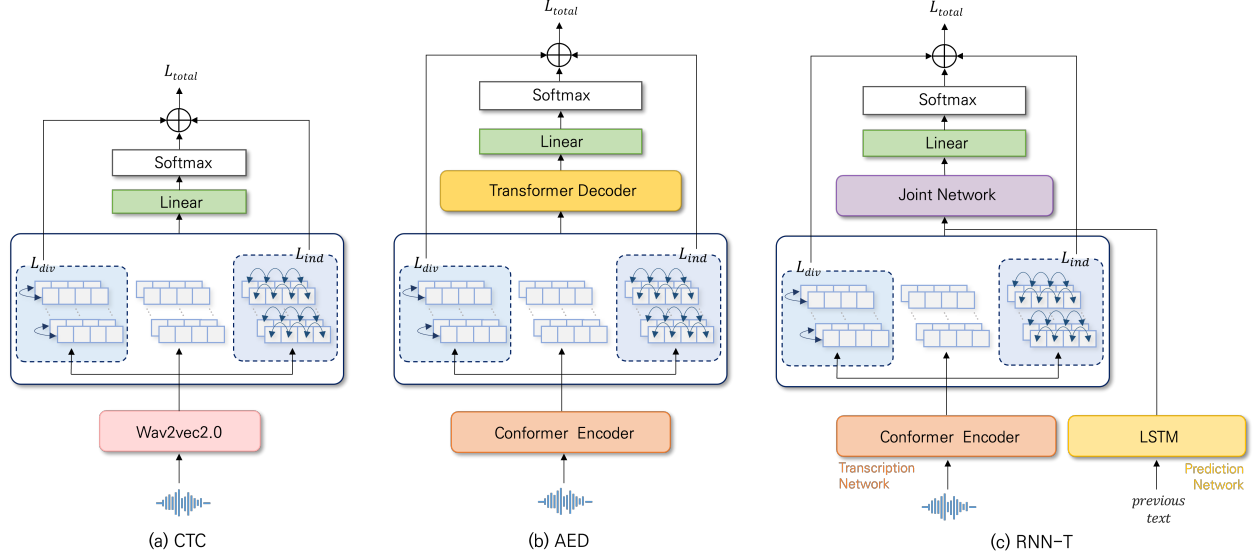


Figure 1: Block schematic of the E2E ASR models incorporating diversity and independence regularization during training.

prediction (DLP) to mitigate deletion and insertion errors in the CTC output. A similar line of work applies auxiliary CTC losses to intermediate layers for performance gains [23]. For AED models, reducing redundancy among character-level feature vectors lower the word error rate (WER) [24]. RNN-T models have also incorporated regularization methods to address sequential alignment [25]. However, most existing regularization techniques are specialized for certain model architectures or masking processes, or focus on controlling feature variance and covariance in self-supervised learning environments.

3. Proposed method

To improve the representational capability of speech recognition models, our regularization approach introduces two loss terms during training—one to increase feature diversity and the other to reduce redundancy across dimensions. This regularization promotes the learning of more expressive representations in feature space, enabling the model to capture a broader range of speech patterns more effectively. The regularization term consists of the diversity loss and the independence loss, which are combined with the ASR loss to define the final objective function. We implement and evaluate our regularization scheme on CTC, AED, and RNN-T models, as illustrated in Figure 1.

3.1. Diversity loss

Speech signals exhibit high correlations between adjacent samples, which limits the representational power of neural networks [26]. Diversity loss reduces similarity among feature vectors within an utterance, mitigating correlation between adjacent samples. This encourages learned representations to capture a broader range of information and perspectives, ultimately enhancing the model’s capacity for nuanced feature extraction. Diversity loss is calculated based on the cosine similarity between pairs of normalized feature vectors c_i and c_j extracted from the encoded feature $C = \{c_1, \dots, c_N\}$, where N denotes the sequence length. Since cosine similarity can be negative, the loss function applies its absolute value. The diversity loss L_{div} is then defined as:

$$L_{div} = \frac{2}{N(N-1)} \sum_{i \neq j} |\cos(c_i, c_j)|, \quad (1)$$

$$\cos(c_i, c_j) = \frac{c_i \cdot c_j}{\|c_i\| \|c_j\|}. \quad (2)$$

The notation $\sum_{i \neq j}$ indicates summation over all pairs i, j where $i \neq j$, with i and j representing distinct indices of feature vectors extracted from a given utterance. $\cos(c_i, c_j)$ represents the cosine similarity between feature vector pairs c_i and c_j , where $c_i \cdot c_j$ is the dot product of vectors c_i and c_j , and $\|c_i\|$ is the norm of vector c_i . To ensure that self-similarity is not penalized, our approach excludes diagonal elements of the cosine similarity matrix when computing diversity loss.

3.2. Independence loss

Independence loss encourages the learned feature vectors to contain independent information. The goal is to reduce the information redundancy by minimizing the covariance between feature vectors. This is achieved by minimizing the off-diagonal elements of the covariance matrix calculated from the feature vectors. Independence loss L_{ind} is defined as follows:

$$L_{ind} = \sum_{i \neq j} (\text{cov}(c_i, c_j))^2, \quad (3)$$

where $\text{cov}(c_i, c_j)$ represents the covariance between feature vectors c_i and c_j , which is calculated as:

$$\text{cov}(c_i, c_j) = \frac{1}{K} \sum_{k=1}^K (c_{ik} - \bar{c}_i)(c_{jk} - \bar{c}_j), \quad (4)$$

where $k \in \{1, \dots, K\}$ represents the dimension index within the feature vector and K represents the total number of dimensions. Additionally, c_{ik} and c_{jk} refer to the k -th elements of the feature vectors c_i and c_j , respectively. The terms $\bar{c}_i$ and $\bar{c}_j$ represent the mean values of the feature vectors c_i and c_j , respectively. Covariance measures the strength of the linear relationship between two variables, with higher values indicating a stronger linear relationship. Therefore, by minimizing the

square of the covariance, the model is encouraged to ensure that the feature vectors contain independent information.

3.3. Final Loss Function

For model optimization, we employ a combined objective function that integrates diversity loss L_{div} and independence loss L_{ind} with the ASR loss L_{ASR} , which varies depending on the E2E model architecture, such as the CTC loss, AED loss, or RNN-T loss. The total loss function L_{total} is defined as follows:

$$L_{\text{total}} = L_{\text{ASR}} + \lambda_{\text{div}} L_{\text{div}} + \lambda_{\text{ind}} L_{\text{ind}}, \quad (5)$$

where λ_{div} and λ_{ind} are the weights for diversity loss and independence loss, respectively. To observe performance changes, we experimented by varying these weights from 0.1 to 1.0.

4. Experiments

4.1. Datasets

We conducted experiments on the LibriSpeech [27], TEDLIUM Release 2 (TEDLIUM2) [28], CHiME-3 [29], and KsponSpeech [30] datasets. The primary experiments for CTC, AED, and RNN-T models were performed on the LibriSpeech training dataset, which contains 960 hours of English speech data. Model evaluation was conducted using the test-clean, and test-other subsets. For the CTC model, additional experiments were conducted on TEDLIUM2, CHiME-3, and KsponSpeech datasets. TEDLIUM2 consists of 207 hours of transcribed TED talk recordings, and we utilized the dev and test sets for evaluation. KsponSpeech provides 965 hours of Korean conversational speech, with evaluation subsets (eval-clean and eval-other) categorized based on perplexity. KsponSpeech is a large-scale Korean conversational dataset that includes repeated words and phonemes, and frequent context shifts. As a result, it inherently exhibits high redundancy among feature vectors within an utterance. CHiME-3 contains noisy speech recorded in various indoor and outdoor environments, with 9,137 training utterances and dt and et subsets provided for evaluation.

4.2. Experimental Setup

Toolkits. We trained the CTC, AED, and RNN-T models using frameworks optimized for each architecture: Fairseq [31] for CTC, ESPnet [32] for AED, and Icefall for RNN-T.

Training details. CTC models were fine-tuned end-to-end using a pretrained wav2vec 2.0 model [33] using the CTC loss and the Adam optimizer with a learning rate of $1e-5$. For AED, we trained models with a conformer-transformer architecture, comprising a 12-block conformer encoder (512 output dimensions, four attention heads, 2048 feed-forward dimensions), along with a 6-block transformer decoder. AED model optimization was performed using the Adam optimizer with a learning rate of 0.0025. RNN-T models were trained using a conformer encoder, an LSTM decoder, and a Noam-scheduler-based optimizer with a learning rate factor of 2.5 for 35 epochs. The best checkpoints for all architectures were selected based on the lowest validation WER.

Evaluation. Model performance was evaluated using WER, and we additionally employed the RERR for comparisons with existing methods. RERR is calculated by comparing the baseline WER with the WER of the proposed method, and then averaging over both development and test sets. All evaluations were performed using viterbi, beam search, and greedy search decoding, with WER measured without language models.

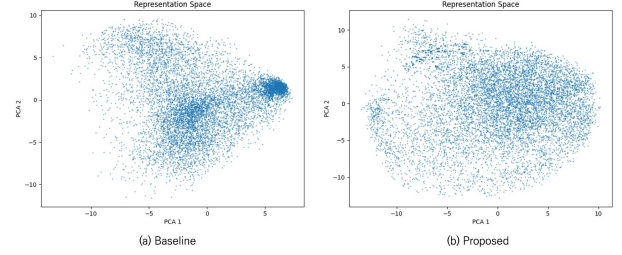


Figure 2: Representation distributions of the baseline CTC model (a) and the proposed model with diversity and independence losses (b) on the CHiME dataset.

Table 1: WER(%) evaluations of CTC, AED, and RNN-T models on the LibriSpeech test-clean and test-other subsets.

Model	Method	λ_{div}	λ_{ind}	LibriSpeech		RERR
				test-clean	test-other	
CTC	Baseline	-	-	4.50	10.52	-
	Proposed	1.0	1.0	4.39	10.33	2.13%
	Proposed	0.1	1.0	4.34	10.21	3.25%
	Proposed	1.0	0.1	4.35	10.29	2.76%
AED	Baseline	-	-	2.75	6.53	-
	Proposed	1.0	1.0	2.60	6.39	3.80%
	Proposed	0.1	1.0	2.65	6.44	2.51%
	Proposed	1.0	0.1	2.63	6.42	3.02%
RNN-T	Baseline	-	-	3.27	7.86	-
	Proposed	1.0	1.0	3.09	7.52	4.96%
	Proposed	0.1	1.0	3.13	7.55	4.11%
	Proposed	1.0	0.1	3.14	7.54	4.02%

5. Results

5.1. Effectiveness in various E2E ASR models

Table 1 presents the WER and RERR for the test-clean and test-other sets of the CTC, AED, and RNN-T models trained on 960 hours of LibriSpeech training dataset. Experiments were conducted with three sets of loss weights: $\lambda_{\text{div}} = 1.0$, $\lambda_{\text{ind}} = 1.0$; $\lambda_{\text{div}} = 0.1$, $\lambda_{\text{ind}} = 1.0$; and $\lambda_{\text{div}} = 1.0$, $\lambda_{\text{ind}} = 0.1$. Across all of the E2E model architectures, our proposed diversity-independence regularization method led to significant WER reductions, with the highest RERR of 4.96% observed in the RNN-T model. CTC and AED models also showed substantial improvements, achieving RERR values between 2.13% and 3.80%, demonstrating the model-agnostic nature of our approach. Furthermore, performance improvements were observed even with a low weight of 0.1 for either λ_{div} or λ_{ind} , indicating that these regularization terms enhance model performance without requiring large weight values. The effectiveness of our approach stems from maximizing entropy of individual data samples, which broadens the representation space and disperses representation vectors, as illustrated in Figure 2.

5.2. Robustness to noisy and out-of-distribution data

To evaluate the robustness and generalization of our method, we conducted experiments on both noisy and out-of-distribution (OOD) datasets. For noise robustness evaluation, we used the KsponSpeech dataset, which consists of spontaneous Korean speech with disfluencies and colloquial expressions. The KsponSpeech results in Table 2 shows that our method consistently outperformed the baseline CTC model across all loss

Table 2: WER(%) results of CTC models on the KsponSpeech evaluation subsets, eval-clean (clean) and eval-other (noisy).

Method	KsponSpeech		RERR
	eval-clean	eval-other	
Baseline	29.33	35.32	-
Proposed ($\lambda_{\text{div}}=1.0$, $\lambda_{\text{ind}}=1.0$)	27.21	33.44	6.28%
Proposed ($\lambda_{\text{div}}=0.1$, $\lambda_{\text{ind}}=1.0$)	27.62	33.61	5.34%
Proposed ($\lambda_{\text{div}}=1.0$, $\lambda_{\text{ind}}=0.1$)	27.91	33.76	4.63%

Table 3: OOD WER(%) evaluation results for CTC models trained on different datasets (CHiME-3, LibriSpeech, TEDLIUM2), with evaluation conducted on CHiME-3 validation sets.

Train Set	Method	CHiME-3		RERR
		dt	et	
CHiME-3	Baseline	13.34	30.63	-
	Proposed ($\lambda_{\text{div}}=1.0$, $\lambda_{\text{ind}}=1.0$)	12.95	30.10	2.33%
	Proposed ($\lambda_{\text{div}}=0.1$, $\lambda_{\text{ind}}=1.0$)	12.96	30.06	2.35%
	Proposed ($\lambda_{\text{div}}=1.0$, $\lambda_{\text{ind}}=0.1$)	13.04	30.17	1.88%
LibriSpeech	Baseline	27.67	52.19	-
	Proposed ($\lambda_{\text{div}}=1.0$, $\lambda_{\text{ind}}=1.0$)	27.21	50.97	2.00%
	Proposed ($\lambda_{\text{div}}=0.1$, $\lambda_{\text{ind}}=1.0$)	27.19	50.94	2.06%
	Proposed ($\lambda_{\text{div}}=1.0$, $\lambda_{\text{ind}}=0.1$)	27.24	51.01	1.91%
TEDLIUM2	Baseline	27.72	58.70	-
	Proposed ($\lambda_{\text{div}}=1.0$, $\lambda_{\text{ind}}=1.0$)	27.64	57.54	1.13%
	Proposed ($\lambda_{\text{div}}=0.1$, $\lambda_{\text{ind}}=1.0$)	27.63	57.52	1.17%
	Proposed ($\lambda_{\text{div}}=1.0$, $\lambda_{\text{ind}}=0.1$)	27.65	57.63	1.04%

Table 4: WER(%) comparison with existing methods using the CTC model, with both training and evaluation conducted on the TEDLIUM2 dataset.

Method	TEDLIUM2		RERR
	dev	test	
Baseline	8.86±0.048	8.51±0.042	-
Mask-CTC [21]*	8.78	8.45	0.80 %
Improved Mask-CTC [22]*	8.72	8.43	1.26 %
Proposed ($\lambda_{\text{div}}=1.0$, $\lambda_{\text{ind}}=1.0$)	8.59±0.009	8.35±0.013	2.46%
Proposed ($\lambda_{\text{div}}=0.1$, $\lambda_{\text{ind}}=1.0$)	8.58±0.028	8.31±0.045	2.76%
Proposed ($\lambda_{\text{div}}=1.0$, $\lambda_{\text{ind}}=0.1$)	8.63±0.015	8.38±0.012	2.06%

* Results reproduced from our experiments.

weight settings, demonstrating its ability to handle noisy conditions effectively. The generalization performance was validated using the CHiME-3 evaluation subsets, where the models were trained on CHiME-3, LibriSpeech, and TEDLIUM2, respectively. As shown in Table 3, our approach consistently yielded RERR improvements across all OOD settings, showcasing its effectiveness in adapting to previously unseen acoustic conditions. Furthermore, since CHiME-3 includes speech recorded in noisy and reverberant environments, further validate the noise robustness of our method.

Table 5: WER(%) evaluation of ablation studies on the CTC model trained and evaluated on the TEDLIUM2 dataset.

Method	TEDLIUM2		RERR
	dev	test	
Baseline	8.86	8.51	-
+ Independence	8.72	8.46	1.08%
+ Diversity	8.69	8.39	1.66%
+ Independence & Diversity	8.58	8.31	2.76%

5.3. Comparison with other regularization methods

We evaluated our method against existing regularization techniques, including Mask-CTC [21] and Improved Mask-CTC [22], using the CTC model on TEDLIUM2. As shown in Table 4, our approach achieved the highest RERR of 2.76%, surpassing Mask-CTC (0.80%) and Improved Mask-CTC (1.26%), confirming its superior effectiveness. To ensure result reliability, we conducted three independent runs and reported 95% confidence intervals in Table 4, providing a statistically robust comparison. These intervals account for variability across runs, confirming that the improvements are not due to chance but represent a meaningful advantage. Moreover, this study is the first to introduce and compare regularization techniques for stabilizing E2E ASR models, laying the groundwork for future research on improving generalization and robustness.

5.4. Ablation studies

Ablation studies on the TEDLIUM2 dataset assessed the individual and combined effects of diversity and independence regularization. Table 5 compares the baseline wav2vec 2.0 CTC model with models applying each regularization method separately and together. Applying independence regularization alone, with $\lambda_{\text{div}}=0.1$ and $\lambda_{\text{ind}}=1.0$, resulted in an RERR of 1.08%, demonstrating its effectiveness in improving model generalization. Similarly, diversity regularization alone led to an RERR of 1.66%, highlighting its role in enhancing feature representations. When both regularization techniques were applied together, the model achieved the best performance, with an RERR of 2.76%. These results indicate that both techniques improve performance independently while offering complementary and orthogonal benefits when combined.

6. Conclusion

In this paper, we proposed a novel regularization approach to improve E2E ASR models by introducing feature diversity and independence losses. Our method enables the model to learn richer and more expressive features, ultimately enhancing recognition accuracy. The proposed regularization effectively mitigates overfitting and significantly improves generalization, contributing to robustness across diverse speech conditions.

7. Acknowledgements

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (RS-2025-00557944)

8. References

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